Macroevolution Unit
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Chloe Nash, PhD

Current Position

2023–Present **Postdoctoral Scholar of Ichthyology**, Macroevolution Unit.
Okinawa Institute for Science and Technology, Japan
PI: Lauren Sallan

Education

2016–2023 **University of Chicago, IL**, *Ph.D.*, Committee on Evolutionary Biology.
Thesis: Phylogeny, Biogeography, and Ecomorphology of the Goatfishes (Family Mullidae)
Advisor: Mark Westneat

2012–2016 **Wesleyan University, CT**, *B.A.*, Biology (Honors) and Environmental Studies.
Advisor: Barry Chernoff

Spring 2015 **James Cook University, Townsville, QLD, Australia**, Marine Biology.
Institute for Study Abroad: Butler University

Research Interests

Phylogenetics of Marine Fishes, Phylogenetic Comparative Methods, Comparative Biogeography, Evolutionary and Functional Dynamics of Marine Fish Communities.

Publications

[1] **Nash, Chloe M.**, Kelly J. Kang^{+}, Lauren Sallan, and Mark W. Westneat. "Bioregionalization identifies phylogenetic and spatial diversity in a globally distributed clade of coral reef fishes". *In Review at Sytematic Biology.*

[2] **Nash, Chloe M.**, Marcus T. Bartholomew^{+}, Akimi Sugiyama^{+}, Roger M. Wilder^{+}, Samuel J. Coatham, and Lauren Sallan. "Strong Taxonomic Turnover but Limited Change in Functional Redundancy Across a Seagrass-to-Coral Gradient". *In Prep.*

[3] **Chloe M. Nash**, Linnea L. Lungstrom, Lily C. Hughes, and Mark W. Westneat. "Phylogenomics and body shape morphometrics reveal recent diversification in the goatfishes (Syngnatharia: Mullidae)". In: *Molecular Phylogenetics and Evolution* 177 (2022), p. 107616. ISSN: 1055-7903. DOI: https://doi.org/10.1016/j.ympev.2022.107616.

[4] Lily C Hughes, **Nash, Chloe M**, William T White, and Mark W Westneat. "Concordance and Discordance in the Phylogenomics of the Wrasses and Parrotfishes (Teleostei: Labridae)". In: *Systematic Biology* (Nov. 2022). ISSN: 1063-5157. DOI: 10.1093/sysbio/syac072.

[5] Charlene L. McCord, **Nash, Chloe M.**, W. James Cooper, and Mark W. Westneat. "Phylogeny of the damselfishes (Pomacentridae) and patterns of asymmetrical diversification in body size and

feeding ecology". In: *PloS one* 16.10 (2021), e0258889. DOI: https://doi.org/10.1371/journal.pone.0258889.

[6] **Nash, Chloe M.**$^{+}$, Michelle L. Kraczkowski, and Barry Chernoff. "Insight into the population structure of hardhead silverside, *Atherinomorus stipes* (Teleostei: Atherinidae), in Belize and the Florida Keys using nd2". In: *Ecology and Evolution* (2017). ISSN: 2045-7758. DOI: 10.1002/ece3.3457.

[7] **Nash, Chloe M.** "Digest: Shifting biomes: Insight into patterns of plant radiation and dispersal". In: *Evolution* 71.8 (2017), pp. 2130–2131. ISSN: 1558-5646. DOI: 10.1111/evo.13300.

[8] H. Barrios-Garrido, K. De Turris-Morales, **C. M. Nash**$^{+}$, G. Delgado-Ortega, and N. Espinoza-Rodríguez. "Acoustic parameters of Guiana dolphin (*Sotalia guianensis*) whistles in the southern Gulf of Venezuela". In: *Aquatic Mammals* 42.2 (2016). ISSN: 01675427. DOI: 10.1578/AM.42.2.2016.127.

[9] **Nash, Chloe M.**$^{+}$. "Island Population Structure of Hardhead Silversides, *Atherinomorus stipes*, in Belize and the Florida Keys". Senior Honors Thesis. Wesleyan University, 2016, pp. 1–56.

Conference Presentations and Posters

[1] **Chloe M. Nash***, Marcus Bartholomew, Akimi Sugiyama, Roger Wilder, and Lauren Sallan. "The Other Blue Zone: Trait diversity and biogeography of coral reef fishes across the Ryukyu Islands". Oral Presentation. *Joint Meeting of Ichthyologists and Herpetologists*, July 8-12, 2026. New Orleans, Louisiana.

[2] **Chloe M. Nash***, Mia Balis, Marcus Bartholomew, Hana Yuki, and Lauren Sallan. "The Other Blue Zone: Trait diversity and biogeography of coral reef fishes across the Ryukyu Islands". Oral Presentation. *2025 Joint Conference of the Asian Society of Ichthyologists Annual Meeting and the 12th Annual Indo-Pacific Fish Conference*, June 9-13, 2025. Taipei, Taiwan.

[3] **Chloe M. Nash***, Samuel Coatham, and Lauren Sallan. "The Other Blue Zone: Trait diversity and biogeography of coral reef fishes across the Ryukyu Islands". Oral Presentation. *Society for Integrative and Comparative Biology*, Jan. 3-7, 2025. Atlanda, GA.

[4] **Chloe M. Nash***, Samuel Coatham, and Lauren Sallan. "The Other Blue Zone: Trait diversity and biogeography of coral reef fishes across the Ryukyu Islands". Oral Presentation. *Okinawa Marine Science Workshop*, Nov. 22, 2024. Okinawa, Japan.

[5] **Chloe M. Nash***, Rikako Ozaki, and Lauren Sallan. "Reef Fish Diversity Across the Ryukyu Island". Poster. *Evolution*, July 26-30, 2024. Montreal, Canada.

[6] **Chloe M. Nash***, Linnea L. Lungstrom^{+}, and Mark W. Westneat. "Evolution of substrate use across global assemblages of the goatfishes (Mullidae)". Oral Presentation. *Society for Integrative and Comparative Biology*, Jan. 2-6, 2024. Seattle, WA.

[7] **Chloe M. Nash***, Linnea L. Lungstrom^{+}, and Mark W. Westneat. "Evolution of substrate use across global assemblages of the goatfishes (Mullidae)". Oral Presentation. *Indo-Pacific Fish Conference*, Nov. 20-24, 2023. Auckland, New Zealand.

[8] **Chloe M. Nash***, Linnea L. Lungstrom^{+}, and Mark W. Westneat. "Evolution of substrate use across global assemblages of the goatfishes (Mullidae)". Oral Presentation. *Evolution*, June 24-28, 2022. Cleveland, OH.

[9] **Chloe M. Nash**[*]. "Comparative biogeographic patterns of coral reef fishes using occurrence-based regionalization methods". Oral Presentation. *Society of Systematic Biology Standalone Meeting*, January 3-6, 2020. Gainesville, FL.

[10] **Chloe M. Nash**[*] and Mark Westneat. "Biogeographic patterns of the damselfishes and clownfishes (Pomacentridae) using occurrence and trait based regionalization methods". Oral Presentation. *Evolution*, June 21-25, 2019. Providence, RI.

[11] **Chloe M. Nash**[*], Andrew B. George, Charlene L. McCord, and Mark W. Westneat. "Functional Biogeography: Patterns of SpatioTemporal Evolution of Biomechanical Traits in the Triggerfishes (Balistidae)". Oral Presentation. *Society of Integrative and Comparative Biology*, January 3-8, 2019. Tampa, FL.

[12] Kirsten Peramba[*], **Chloe Nash**, Dahila Walters[+], Lilly Hackworth[+], Matthew Schumm[+], Oscar Pineda, and Eric Edsinger. "Squid Monday: Characterizing population structure in *Doryteuthis pealeii*". Poster. *Society of Integrative and Comparative Biology*, January 3-8, 2019. Tampa, FL.

[13] **Chloe M. Nash**[*]. "FishLife: Biogeography". Tutorial. *FishLife Collaborators Meeting*, October 5-7, 2018. Chicago, IL.

[14] **Chloe M. Nash**[*], Charlene L. McCord, and Mark Westneat. "Biogeography of the Pomacentridae with Insight into the Coral Triangle". Poster. *2018 Society of Systematic Biologists Standalone Meeting*, June 1-4, 2018. Columbus, OH.

[15] **Chloe M. Nash**[*], Charlene L. McCord, and Mark Westneat. "Biogeography of the Pomacentridae with Insight into the Coral Triangle". Oral Presentation. *Indo-Pacific Fish Conference*, October 2-6,2017. Tahiti, French Polynesia.

[16] Kayla Anatone[*], Barry Chernoff, Michelle Kraczkowski, Abrial Meyer[+], and **Chloe Nash**[+]. "Barriers to Gene Flow for Two Species of Dace *Rhinichthys* in Three Connecticut Streams". Oral Presentation. *Joint Meeting of Ichthyologists and Herpetologists*, July 12-16, 2017. Austin, TX.

[17] **Chloe M. Nash**[*+], Michelle L. Kraczkowski, and Barry Chernoff. "Genetic Structure of *Atherinomorus stipes*, Hardhead Silverside, in Belize and Florida based on *nd2*". Oral Presentation. *Joint Meeting of Ichthyologists and Herpetologists*, July 6-10, 2016. New Orleans, LA.

[18] Kayla Anatone[*], Barry Chernoff, Michelle Kraczkowski, Abrial Meyer[+], Nicole Roman-Johnston[+], **Chloe Nash**[+], Julio Angel[+], Alexandra Fireman[+], and Cindy Flores[+]. "The Genetic Structure of Longnose Dace, *Rhinichthys cataractae*, from Two Waterfalls in Connecticut". Poster. *Joint Meeting of Ichthyologists and Herpetologists*, July 6-10, 2016. New Orleans, LA.

[19] **Chloe M. Nash**[*+], Michelle Kraczkowski, and Barry Chernoff. "Island Population Structure of Hardhead Silversides, *Atherinomorus stipes*, in Belize". Poster. *Joint Meeting of Ichthyologists and Herpetologists*, July 15-19, 2015. Reno, NV.

[20] Nicole Roman-Johnston[+], Kayla Anatone[*], Julio Angel[+], Alexandra Fireman[+], Abrial Meyer[+], **Chloe Nash**[+], Michelle Kraczkowski, and Barry Chernoff. "Over the Falls: the Effect of Waterfalls on the Genetic Structure of the Eastern Blacknose Dace, *Rhinichthys atratulus*, in Connecticut". Poster. *Joint Meeting of Ichthyologists and Herpetologists*, July 15-19, 2015. Reno, NV.

[21] Barry Chernoff[*], Michelle Kraczkowski, Julio Angel[+], Abrial Meyer[+], **Chloe Nash**[+], Nicole Roman-Johnston[+], and Sophie Breitbart[+]. "Over the falls: Population Differentiation of Blacknose and Longnose Dace, *Rhinichthys atratulus* and *R. cataractae*, in Connecticut". Oral Presentation. *Joint Meeting of Ichthyologists and Herpetologists*, July 30-Aug 3, 2014. Chattanooga, TN.

Mentored Student Presentations

[1] Dominik Kopcak*+, Jonathan Chang, **Chloe M. Nash**., Isaac Trinidade Santos, Michael Alfaro, and Lauren Sallan. "Diversity imbalance is pervasive across vertebrate trees". Poster. *Evolution Conference*, July 26-30, 2024. Montreal, CA.

[2] Olivia Grobmyer*+, **Chloe M. Nash**., and Mark W. Westneat. "Quantifying Goatfish Feeding Patterns over Rocky and Sandy Substrates". Poster. *Society for Integrative and Comparative Biology*, Jan. 2-6, 2024. Seattle, WA.

[3] Adera Craig*+, **Chloe M. Nash**., and Mark W. Westneat. "Biogeographic and evolutionary trends of feeding specializations in butterflyfishes (Chaetodontidae)". Poster. *Society for Integrative and Comparative Biology*, Jan. 2-6, 2024. Seattle, WA.

[4] Lucia Reyes*+, **Chloe Nash**, and Lauren Sallan. "REPRODUCTION PATTERNS OF MARINE FISHES FROM LARGE MARINE ECOSYSTEMS (LMEs) IN JAPAN". Poster. *Board of Governors Meeting, OIST*, Sept. 27, 2023. Okinawa, Japan.

[5] Olivia Grobmyer*+, **Chloe M. Nash**., and Mark W. Westneat. "Quantifying Goatfish Feeding Patterns over Rocky and Sandy Terrain". Poster. *The 2023 University of Chicago Undergraduate Research Symposium*. Chicago, IL.

[6] Adera Craig*+, **Chloe M. Nash**., and Mark W. Westneat. "Potential Impacts of Climate Change on Butterflyfish: A Phyloclimatic and Biogeographic Modeling of Chaetodon". Poster. *The 2023 University of Chicago Undergraduate Research Symposium*. Chicago, IL.

[7] Linnea L. Lungstrom*+, **Chloe M. Nash**, and Mark W. Westneat. "Geometric morphometrics of the goatfishes (Mullidae) to explore ecomorphological patterns". Poster. *Society of Integrative and Comparative Biologists*, January-March, 2021. Virtual.

[8] Kelly Kang*+, **Chloe M. Nash**, Andrew B. George, and Mark W. Westneat. "Comparative biogeography and geometric morphometrics of the balistoid fishes". Poster. *Society of Integrative and Comparative Biologists*, January-March, 2021. Virtual.

[9] Linnea L. Lungstrom*+, **Chloe M. Nash**., and Mark W. Westneat. "Ecomorphological Relationships of the Goatfishes (Family Mullidae)". Poster. *The 2020 University of Chicago Undergraduate Research Symposium Proceedings*. Virtual.

[10] Kelly J. Kang*+, **Chloe M. Nash**., and Mark W. Westneat. "Comparative Biogeography of the Filefishes (Family Monacanthidae)". Poster. *The 2020 University of Chicago Undergraduate Research Symposium Proceedings*. Virtual.

* Presenter + Undergraduate/Rotation Student

Awards and Grants

2026 **COI-NEXT**, OIST, *¥3,000,000*, with H. Hata and L. Sallan

2019 **Edward C. Raney Award**, American Society of Ichthyologists and Herpetologists, *$1,000*

2019 **France And Chicago Collaborating in The Sciences Grant**, France Chicago Center, *$21,900*, with A. George and M. Westneat

2019 **Hinds Fund Award**, University of Chicago, *$2,500*

2018 **Graduate Student Research Award**, Society of Systematic Biology, *$2,000*

2018 **Steiner Award**, University of Chicago, *$1,500*

2018 **Lerner-Gray Memorial Fund**, American Museum of Natural History, *$2,485*

2018 **CEB Summer Travel Award**, University of Chicago, *$2,000*

2018 **NSF Graduate Research Fellowship**, *Honorable Mention*

2017 **Graduate Council Student Travel Grant**, University of Chicago, *$600*

2017 **IPFC Student Housing Grant**, *equivalent to $250*

2015 **College of the Environment Internship**, Wesleyan University, *$4,000*

2015 **Valiant Volunteer: 1st Place**, IFSA-Butler Australia, *$100*

2014 **College of the Environment Internship**, Wesleyan University, *$5,000*

2013 **Presidential Fellow of the College of the Environment**, Wesleyan University, *$5,000*

Relevant Skills

Diving, *AAUS Scientific Diver, NAUI Divemaster, over 500 logged dives.*

Coding, *Proficient in R, Python, Bash, Latex.*

Molecular Techniques, *DNA extraction, library preparation, PCR, gel electrophoresis.*

Phylogenomic assembly and analyses, *Whole genome assembly and annotation, SNP analyses, phyluce, IQtree, BEAST, Mr. Bayes.*

Phylogenetic Comparative Methods, *trait/range/composition evolution, lineage diversification analyses.*

Geometric Morphometric Analyses, *Land-marking, PCA, comparative analyses.*

Occurrence based Biogeography, *GBIF, VertNet, OBIS.*

Museum Collection, *Specimen collection, preparation, fixation, tissue preparation, cataloging.*

Field Work

2023-Present **Ryukyu Islands Connectivity**, Japan

July 2021 **Mullidae Behavior and Kinematics**, Mo'orea, French Polynesia

Summer 2019 **Mullidae Behavior and Kinematics**, Mo'orea, French Polynesia

Oct. 2017 **Mullidae Field Pilot Project**, Mo'orea, French Polynesia

July 2015 ***Atherinomorus stipes* Collection**, Florida Keys

May 2015 **Flatback Tagging Project Release**, Whitsundays, Queensland, Australia

July 2014 ***Atherinomorus stipes* Collection**, Carrie Bow Cay, Belize

Summer 2013 **Great White Shark Dorsal Fin ID**, Mossel Bay, South Africa

Invited Talks and Symposia

2022 **"It Depends" Symposium**, *Evolution Conference*

2021 **Marine Biology Guest Lecture**, *University of St. Josephs*

2020 **Fish Group**, *University of Chicago*

2019 **Fish Group**, *University of Chicago*

Workshops

June. 2026 **Applying Theories of How We Learn**, *OIST*, Okinawa, JP

Jan. 2026 **RevBayes Workshop**, *Society of Systematic Biologists*, Baton Rouge, LA

Mar. 2024 **SQuiD Statistics Workshop**, *University of Hokkaido*, Sapporo, JP

Dec. 2023 **Pathways Toward an Integrated Understanding of the Marine Environment and Marine Ecosystems**, *Tohoku University*, Sendai, JP

Oct. 2023 **Evolutionary Morphology Workshop**, *OIST*, Okinawa, JP

Jan. 2020 **Future of Systematics**, *Society of Systematic Biologists*, Gainesville, FL

Jan. 2020 **Software School for Phylogenomics**, *Society of Systematic Biologists*, Gainesville, FL

June 2019 **Midwest Phylogenetics Workshop**, Itasca Field Station, MN

Feb. 2019 **Course Design for STEM Professors or Outreach Professionals**, *CCT*,Chicago, IL

Oct. 2018 **FishLife Collaborators Meeting**, *FishLife*,Chicago, IL

Sept. 2018 **Teaching@Chicago**, *UChicago Center for Teaching*,Chicago, IL

May 2018 **Predictive Phylogeography**, *Society of Systematic Biologists*, Colombus, OH

May 2017 **State of the Tree of Life**, *FuturePhy*, Field Museum, Chicago, IL

Oct. 2016 **Fishes Working Group**, *FuturePhy*, Field Museum, Chicago, IL

July 2016 **Introduction to R**, *American Society of Ichthyologists and Herpetologists*, New Orleans, LA

Teaching

Fall 2026 **TA**, *Introduction to Scientific Diving*, OIST

Fall 2022 **TA**, *Marine Biodiversity and Genomics (BIOS27723)*, Marine Biological Laboratory

Fall 2021 **TA**, *Marine Biodiversity and Genomics (BIOS27723)*, Marine Biological Laboratory

Winter 2021 **TA**, *Biomechanics (BIOS22245)*, University of Chicago

Fall 2019 **TA**, *Marine Biodiversity and Genomics (BIOS27723)*, Marine Biological Laboratory

Winter 2019 **TA**, *Animal Behavior (BIOS23249)*, University of Chicago

Fall 2018 **TA**, *Marine Biodiversity and Genomics (BIOS27723)*, Marine Biological Laboratory

Fall 2015 **Organizer/Instructor**, *Marine Biology Student Forum (ENVS419.04)*, Wesleyan University

Fall 2013 **Biology Tutor**, *Principles of Biology I (BIOS181)*, Wesleyan University

Summer 2013 **Marine Education Intern**, *Shark Biology and Conservation*, Mossel Bay, South Africa

2010–2012 **High School Tutor**, *Biology, Geometry, and Algebra*, Baton Rouge, LA

* TA = Teaching Assistant

Undergraduates, Interns, and Rotation Students Mentored

2025-2026 Roger Wilder, OIST, now PhD Student at Texas Tech

2025-2026 Akimi Sugiyama, OIST, undergraduate at Tohoku University

2025-2026 Jamie Suesser, OIST/Northeastern Marine Biology MSc Program

2025 Nicole Yap, OIST Rotation Student

2024-2025 Mia Balis, OIST, undergraduate at Northeastern University
2024-2025 Hana Yuki Burdge, OIST
2024-2025 Leighna Sugimoto, OIST, now MSc Student at James Cook University, AUS
2024-2025 Marcus Bartholomew, OIST/Northeastern Marine Biology MSc Program
2024 Riko Chinen, OIST Rotation Student
2024 Dominik Kopcak, OIST Rotation Student
2024 Ian Johnson, OIST Rotation Student
2023-2024 Sophia Pierucci, OIST
2023-2024 Yuxin Liu, OIST
2023-2024 Rikako Ozaki, OIST, now OIST PhD Student
2023-2024 Lucia Reyes, OIST, now in Erasmus Program
2022-2023 Adera Craig, University of Chicago
2022-2023 Olivia Grobmyer, University of Chicago
2019-2021 Linnea Lungstrom, University of Chicago, now PhD Student at UChicago
2019-2023 Kelly Kang, University of Chicago
2015-2016 Sage Loomis, Wesleyan University

Academic Service

2026- Co-Organizer "Marine and Terrestrial Science Section Conference", *OIST*
2021-2022 Co-Organizer "It Depends" Symposium, *Evolution Conference*
2021-2023 BSD DEIship Committee, *University of Chicago*
2021 CEB Admissions Committee, *University of Chicago*
2020- CEB DEI Committee, *University of Chicago*
2020- BSD Graduate Eduction Advisory Committee, *University of Chicago*
2016-2017 TG Student Seminar Series Organizer, *University of Chicago*

Outreach and Volunteer Work

March 2019 Workshop Leader, *Expanding Your Horizons*
2018-2023 Dozin' with the Dinos, *Field Museum of Natural History*
2017-2018 Facilities Committee Co-Chair, *Expanding Your Horizons*
2017 Tween Day Organizer, *Field Museum of Natural History*
2017-2023 Member's Night, *Field Museum of Natural History*
June 2017 Horizon Science Academy Belmont: Biodiversity Project Judge, *Field Museum of Natural History*
March 2017 Buddy, *Expanding Your Horizons*
Feb. 2017 Northeastern Illinois University Student Tour and Panel, *Field Museum of Natural History*
Fall 2015 Wesleyan Science Outreach, *Wesleyan University*

Society Membership

2015–Present	American Society of Ichthyologists and Herpetologists
2016–Present	Society of Systematic Biologists
2016–Present	Field Museum Women in Science
2017–Present	Society for the Study of Evolution
2017–Present	Association for Women in Science
2018–Present	Animal Behavior Society
2018–Present	Society of Integrative and Comparative Biology

References

Dr. Mark Westneat
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Professor in Organismal Biology and Anatomy & Committee on Evolutionary Biology
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Dr. Michael Coates
Dissertation Committee Chair and Graduate Department Chair
Professor in Organismal Biology and Anatomy & Committee on Evolutionary Biology
University of Chicago
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